



1.6.7 Wrap-Up: Cool Down

1. Circle the expressions equivalent to $(25^3)^{\frac{1}{2}}$.

$$(25)^{\frac{3}{2}} \quad (\sqrt{25})^3 \quad (25)^{\frac{2}{3}} \quad (25^{\frac{1}{3}})^2 \quad \sqrt{25^3}$$

Unit 1, Lesson 7: Evaluating Exponential Expressions



Benchmark

MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.



Learning Targets

- ☐ I can evaluate numerical expressions involving rational exponents.
- ☐ I can generate equivalent numerical expressions involving rational exponents.



1.7.1 Warm-Up: Perfect Squares Puzzle

- Complete the table by finding the square of each whole number.

Whole Number	Square	Whole Number	Square	Whole Number	Square
1	$1^2 = 1$	8		15	
2	$2^2 = 4$	9		16	
3	$3^2 = 9$	10		17	
4		11		18	
5		12		19	
6		13		20	
7		14		21	

- Place a digit in each blue box so that the principal square root is a whole number, and the expression is as close to 0 as possible. Use only the digits 1 to 9, at most one time each.
Reminder: The principal square root is the positive square root.

$$\sqrt{\square\square\square} - \sqrt{\square\square} - \sqrt{\square}$$



1.7.2 Collaboration: Simplifying Expressions with Exponents

- With a partner, write the expression $(27)^{\frac{2}{3}}$ in three equivalent forms.

$(27)^{\frac{2}{3}}$			
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- Explain which equivalent form makes it easiest to find the value of $(27)^{\frac{2}{3}}$ without a calculator.

- Find the value of $(27)^{\frac{2}{3}}$ without a calculator.

- Verify the value of $(27)^{\frac{2}{3}}$ using your calculator.

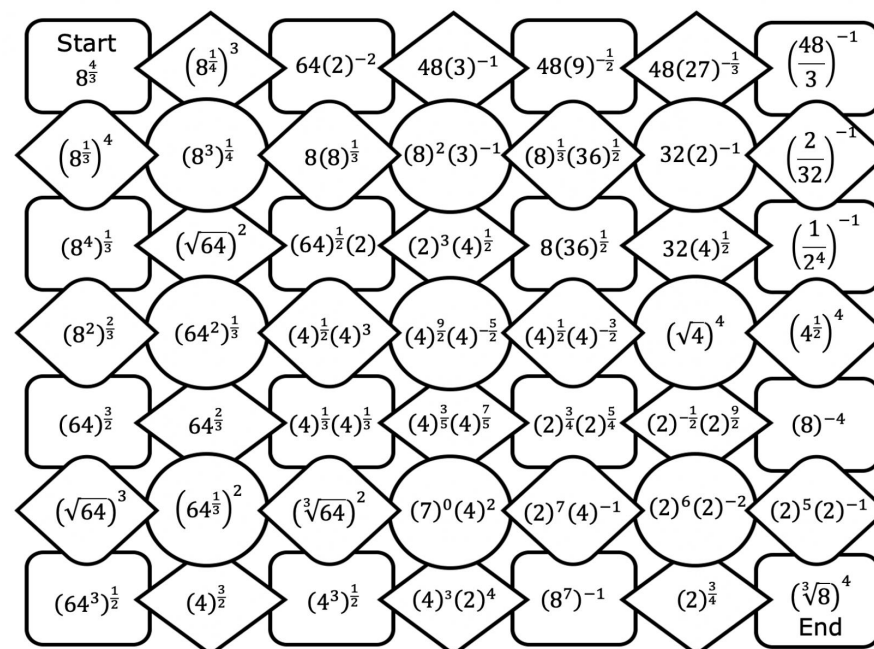
5. For each expression, write an equivalent radical expression. Then, evaluate the expression using your calculator.

Expression	Equivalent Radical Expression	Value of the Expression
$(144)^{-\frac{3}{2}}$		
$-(-64)^{\frac{1}{3}}$		
$(225)^{-\frac{1}{2}}$		
$-9^{\frac{5}{2}}$		
$(\frac{81}{256})^{\frac{3}{4}}$		



1.7.3 Activity: Exponents Maze

1. Work with a partner to complete the maze. Beginning at the "start" box, connect each expression to an equivalent expression to form one continuous path to the "end" of the maze. Move only horizontally or vertically.



2. What is the value of the expression in each box that created the path through the maze?



1.7.4 Your Turn!

1. Evaluate each expression without a calculator.

$(100)^{\frac{1}{2}} \cdot (2^3)$	$(8^{-1}) \left(4^{\frac{5}{2}}\right)$
$(81^6)^{\frac{1}{24}}$	$(125)^{\frac{2}{3}}$
$\left(49^{\frac{1}{2}}\right) (3^{-2})$	$-(3^{12})^{\frac{1}{4}}$
$(2^4) \left(8^{-\frac{2}{3}}\right)$	$\left(\frac{15}{45}\right)^{-4}$
$-\left(\frac{3}{48}\right)^{\frac{3}{2}}$	$\left(\frac{144}{169}\right)^{-\frac{1}{2}}$



1.7.5 Wrap-Up: Reflection

Complete the following statements about today's lesson.

1. Today I am still unsure about ...

2. To fill in this gap I intend to ...